

Practical Session 3

Task 1

Develop a C program using `fork()` that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
*** exited with 0 ***
j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/process_states.c

j_rakshitha@RAKSHITHA:~/ossp_project$ gcc -Wall -Wextra -g src/process_states.c -o
build/process_states
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/process_states
Process State Demonstration

Parent Process Information
PID : 4657
PPID : 941
State: Running

Parent created child with PID: 4658
Parent is waiting for child...

Child Process Information
PID : 4658
PPID : 4657
State: Running

Child is running...
Child is terminating.
Child has terminated.
Parent is terminating.
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

```
GNU nano 8.7.1 src/process_states.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

void display_process_info(const char *name)
{
    printf("%s Process Information\n", name);
    printf("PID : %d\n", getpid());
    printf("PPID : %d\n", getppid());
    printf("State: Running\n\n");
}

int main()
{
    printf("Process State Demonstration\n\n");

    display_process_info("Parent");

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        display_process_info("Child");

        printf("Child is running...\n");
        sleep(5);
        printf("Child is terminating.\n");

        exit(0);
    }
    else
    {
        printf("Parent created child with PID: %d\n", pid);
        printf("Parent is waiting for child...\n\n");

        wait(NULL);

        printf("Child has terminated.\n");
        printf("Parent is terminating.\n");
    }

    return 0;
}

␣␣ Help ␣␣ Write Out ␣␣ Where Is ␣␣ Cut ␣␣ Execute ␣␣ Location ␣␣ Undo
```

Task 2

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
j_rakshitha@RAKSHITHA:~/ossp_project$ top
top - 05:57:03 up 5:41, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7794.6 total, 7074.4 free, 588.1 used, 289.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7206.6 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24456	15468	11556	S	0.0	0.2	0:01.99	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.15	init-syste+
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50404	17156	15832	S	0.0	0.2	0:00.85	systemd-jo+
82	systemd+	20	0	22416	14484	11988	S	0.0	0.2	0:00.16	systemd-re+
92	root	20	0	35380	12332	9228	S	0.0	0.2	0:07.35	systemd-ud+
108	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.06	chronyd-st+
109	root	20	0	4472	3120	2872	S	0.0	0.0	0:00.06	cron
110	message+	20	0	8816	5444	4712	S	0.0	0.1	0:00.25	dbus-daemon
157	root	20	0	44096	29816	14676	S	0.0	0.4	0:00.29	networkd-d+
168	root	20	0	18436	9416	8168	S	0.0	0.1	0:00.21	systemd-lo+
201	syslog	20	0	220548	5776	4464	S	0.0	0.1	0:00.21	rsyslogd
238	root	20	0	5492	2696	2508	S	0.0	0.0	0:00.00	agetty
242	root	20	0	123460	32700	18092	S	0.0	0.4	0:00.21	unattended+
272	_chrony	20	0	23820	11388	7212	S	0.0	0.1	0:00.71	chronyd
282	_chrony	20	0	12096	2468	1820	S	0.0	0.0	0:00.05	chronyd
329	root	20	0	8652	5548	4704	S	0.0	0.1	0:00.01	login
383	j_raksh+	20	0	22340	13584	10968	S	0.0	0.2	0:00.18	systemd
385	j_raksh+	20	0	22912	4108	2108	S	0.0	0.1	0:00.00	(sd-pam)
413	j_raksh+	20	0	6136	5484	3928	S	0.0	0.1	0:00.04	bash
939	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	SessionLea+
940	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.81	Relay(941)
941	j_raksh+	20	0	6268	5544	3872	S	0.0	0.1	0:00.38	bash
3760	polkitd	20	0	306444	8472	7660	S	0.0	0.1	0:00.19	polkitd
4678	j_raksh+	20	0	8180	6016	3900	R	0.0	0.1	0:00.01	top

```
j_rakshitha@RAKSHITHA:~/ossp_project$ ps -eo pid,ppid,state,comm
```

PID	PPID	S	COMMAND
1	0	S	systemd
2	1	S	init-systemd(Ub
7	2	S	init
47	1	S	systemd-journal
82	1	S	systemd-resolve
92	1	S	systemd-udev
108	1	S	chronyd-starter
109	1	S	cron
110	1	S	dbus-daemon
157	1	S	networkd-dispat
168	1	S	systemd-logind
201	1	S	rsyslogd
238	1	S	agetty
242	1	S	unattended-upgr
272	108	S	chronyd
282	272	S	chronyd
329	2	S	login
383	1	S	systemd
385	383	S	(sd-pam)
413	329	S	bash
939	2	S	SessionLeader
940	939	S	Relay(941)
941	940	S	bash
3760	1	S	polkitd
4688	941	R	ps

```
j_rakshitha@RAKSHITHA:~/ossp_project$ ps -eo pid,ppid,state,comm | grep ' R '
```

4698	941	R	ps
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```
j_rakshitha@RAKSHITHA:~/ossp_project$ ps -eo pid,ppid,state,comm | grep ' S '
```

PID	PPID	S	COMMAND
1	0	S	systemd
2	1	S	init-systemd(Ub
7	2	S	init
47	1	S	systemd-journal
82	1	S	systemd-resolve
92	1	S	systemd-udev
108	1	S	chronyd-starter
109	1	S	cron
110	1	S	dbus-daemon
157	1	S	networkd-dispat
168	1	S	systemd-logind
201	1	S	rsyslogd
238	1	S	agetty
242	1	S	unattended-upgr
272	108	S	chronyd
282	272	S	chronyd
329	2	S	login
383	1	S	systemd
385	383	S	(sd-pam)
413	329	S	bash
939	2	S	SessionLeader
940	939	S	Relay(941)
941	940	S	bash
3760	1	S	polkitd
4712	941	S	grep

```
j_rakshitha@RAKSHITHA:~/ossp_project$ ps -eo pid,ppid,state,comm
  PID      PPID S  COMMAND
    1         0 S  systemd
    2         1 S  init-systemd(Ub
    7         2 S  init
   47         1 S  systemd-journal
   82         1 S  systemd-resolve
   92         1 S  systemd-udevd
  108         1 S  chronyd-starter
  109         1 S  cron
  110         1 S  dbus-daemon
  157         1 S  networkd-dispat
  168         1 S  systemd-logind
  201         1 S  rsyslogd
  238         1 S  agetty
  242         1 S  unattended-upgr
  272        108 S  chronyd
  282        272 S  chronyd
  329         2 S  login
  383         1 S  systemd
  385        383 S  (sd-pam)
  413        329 S  bash
  939         2 S  SessionLeader
  940        939 S  Relay(941)
  941        940 S  bash
 3760         1 S  polkitd
 4717         1 S  wsl-pro-service
 4743         92 S  (udev-worker)
 4744         92 S  (udev-worker)
 4750        941 R  ps
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```